Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/01/2024
PI Name:	Suman, Shubhankar Ph.D.		
Project Title:	Senescent Cell Targeting to Alleviate Space Radiation-Induced Neuroinflammation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **CNS**:Risk of Acute (In-flight) and Late Central Nervous System Effects from Radiation Exposure		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2023 HERO NNJ23ZSA001N-OMNIBUS : NASA Human Research Program Omnibus Opportunity
Start Date:	01/01/2024	**End Date:**	12/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2025 per NSSC information (Ed., 12/30/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Li, James Ph.D. (Georgetown University)		
Grant/Contract No.:	80NSSC24K0287		
Performance Goal No.:			
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Task Description:	The cerebral cortex, the outermost layer of the mammalian brain, regulates important functions such as awareness, motor and sensory functions, memory, language, and visual perceptions. Earlier studies using animal models have reported that exposure to heavy-ion space radiation triggered persistent DNA damage and neuroinflammation in the cerebral cortex. However, we have yet to clarify how persistent DNA damage and neuroinflammation are sustained long-term after space radiation exposure. Previous research has linked cellular senescence, the acquisition of the senescence-associated secretory phenotype (SASP), and increased senescence induced inflammatory response (SIR) to the onset of neuroinflammation during aging; however, the role of senescent/SASP cells in chronic neuroinflammation after heavy-ion space radiation warrants further investigation. Our central hypothesis is that persistent DNA damage related to senescence and the senescence inflammatory response (SIR) is the primary driver of sustained neuroinflammatory stress in the cerebral cortex following exposure to space radiation. This investigation aims to identify, quantify, and characterize senescence-susceptible cells in the cerebral cortex and evaluate the efficacy of senescent cell ablation (senolysis) as a strategy to mitigate energetic heavy ions (HZE) or simulated galactic cosmic radiation (GCRsim)-induced neuroinflammation. We propose that targeted removal of these senescent cells may reduce the risk of neuroinflammation. This proposal will expand NASA's research portfolio to encompass the implications of senescence and SIR in relation to space radiation-induced CNS effects. Since many senolytic drugs struggle to penetrate the blood-brain barrier, our approach of genetic senescent cell ablation will provide robust proof-of-principle for the feasibility of senolytic strategies to mitigate HZE-induced CNS effects. This study will enhance our understanding of cell-type-specific biological responses in the cerebral cortex and their role in sustained neuroinflammation. By establishing senescence/SIR signaling as a critical contributor to space radiation-induced neuroinflammation, our investigation will lay the groundwork for future interventions, ultimately aiming to alleviate space radiation-induced CNS effects. Aim 1: Identify, quantify, and characterize senescence/SIR-susceptible cerebral cortex cell types. We will analyze the dose- and time-dependent effects of HZE and gamma radiation on oxidative stress, senescence, SIR, and neuroinflammation. Our working hypothesis posits that the high-LET characteristics of 28Si ions will induce a greater activation of senescence/SIR signaling compared to gamma rays. Aim 2: Obtain "proof-of-principle" that senescent cell removal could alleviate HZE-induced persistent neuroinflammation in the cerebral cortex. Our hypothesis is that the selective removal of senescent cells will lead to a reduction in HZE-induced neuroinflammation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research project aims to advance our understanding of the long-term effects of energetic heavy ions (HZE) and simulated GCR exposures on the CNS, particularly in relation to neuroinflammation and cellular senescence. By identifying the cellular mechanisms that sustain neuroinflammatory stress in the cerebral cortex, we will provide critical insights into the fundamental processes of aging and age-related neurological diseases. Implications of our findings extend beyond space exploration. As populations age on Earth, neuroinflammatory conditions, including Alzheimer's disease and other forms of dementia, pose significant public health challenges. The identification of senescence and the senescence inflammatory response (SIR) as pivotal contributors to sustained neuroinflammation could lead to novel therapeutic strategies that target senescent cells, potentially reducing the burden of these age-associated diseases. Moreover, the exploration of senolytic therapies may pave the way for innovative treatments that enhance brain health and longevity. By providing a proof of principle for senescent cell ablation strategies, our research could catalyze the development of new drugs or interventions aimed at mitigating neuroinflammation and improving cognitive function in aging populations. In summary, this research not only seeks to protect astronauts from the detrimental effects of space radiation but also holds the potential to significantly impact public health by offering new avenues for the prevention and treatment of neurodegenerative diseases on Earth.
Task Progress:	1) Archived formalin-fixed and flash-frozen tissues from male INK-ATTAC mice were used, and new samples were collected for fisetin (a senolytic agent) countermeasure experiments as part of an ongoing NASA's Human Research Program (HRP) funded project. All samples were processed with corresponding controls and an appropriate sample size for the evaluation of senescence, SASP, and neuroinflammatory markers. 2) We successfully conducted experiments using male INK-ATTAC mice to evaluate a genetic senolytic strategy targeting neuroinflammation and cellular senescence following exposure to space-relevant heavy-ion silicon (28Si 50 cGy) radiation. We observed significant increases in GLB1-positive senescent cells and GFAP-positive neuroinflammatory markers in irradiated mice compared to sham controls. Administered AP20187, an apoptosis inducer, to selectively induce apoptosis in p16Ink4a-expressing senescent cells four and a half months post-radiation and demonstrated that treatment with AP20187 effectively reduced the presence of senescent cells and neuroinflammatory markers, supporting the potential of targeting senescent cells to mitigate radiation-induced effects. 3) Using MRI/MRS based approach and live female C57BL6 mice exposed to simulated galactic cosmic radiation (GCRsim; 75 cGy), we revealed alterations in brain metabolites, with increased lipid and Glx/GABA levels in GCRsim-exposed mice compared to controls. 4) We investigated the efficacy of the pharmacological senolytic agent (Fisetin) in alleviating cellular senescence and neuroinflammation in female C57BL6 mice subjected to GCRsim (75 cGy). Using histological assessments, we noted substantial increases in GLB1 (senescence) and IL-6 (SASP) markers in the brain cortex of GCRsim-exposed mice, which were significantly reduced following Fisetin treatment. 5) We also documented elevated glial fibrillary acidic protein (GFAP) expression and amyloid-beta (Aß 1-42) accumulation in GCRsim-exposed mice, with Fisetin treatment effectively mitigating these changes, suggesting that Fisetin can attenuate neuroinflammation and early amyloid-beta deposition, indicating its potential as a therapeutic agent to protect astronaut health during deep-space missions and to address age-related neurodegenerative diseases. Findings from both genetic and pharmacological approaches provide valuable insights into the role of cellular senescence in neuroinflammation associated with HZE and GCRsim exposure.
Bibliography Type:	Description: (Last Updated: 11/12/2025)
Articles in Peer-reviewed Journals	Kumar K, Fornace AJ Jr, Suman S. "8-OxodG: A potential biomarker for chronic oxidative stress induced by high-LET radiation." DNA (Basel). 2024 Aug 1;4(3):221–38. https://doi.org/10.3390/dna4030015 ; PubMed PMID: 39268222; PubMed Central PMCID: PMC11391509 , Aug-2024

Articles in Peer-reviewed Journals	Suman S. "Enteric nervous system alterations in inflammatory bowel disease: Perspectives and implications." Gastrointest Disord (Basel). 2024 Mar 28;6(2):368-79. https://doi.org/10.3390/gidisord6020025 ; PubMed PMID: 38872954; PubMed Central PMCID: PMC11175598 , Mar-2024
Papers from Meeting Proceedings	Kumar K, Angdisen J, Ma J, Fornace Jr AJ, Suman S. "Testing genetic and pharmacological senolytic approach to alleviate space radiation-induced neuroinflammation in mice." Annual Radiation Research Society (RRS) meeting, Tucson, Arizona, September 15-18, 2024. Abstracts. Annual Radiation Research Society (RRS) meeting, Tucson, Arizona, September 15-18, 2024. , Sep-2024